Lintao Peng

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Education

Nanyang Technological University **Singapore**

- **Joint Ph.D. Student** of School of Electrical and Electronic Engineering 2024.10 - 2025.10
- Supervised by Prof. Cuong Dang

Beijing Institute of Technology **Beijing, China**

- **Ph.D. Student** of Information and Communication Engineering 2020.09 - 2026.06
- Co-supervised by Prof. Jun Zhang (Member of Chinese Academy of Engineering) and Dr. Liheng Bian

Xidian University **Xi'an, Shaanxi, China**

- **B.S.** in Computer Science and Technology 2016.09 - 2020.06

Research Interest

My current research interests are mainly about **computational imaging and sensing**. Specifically, I focus on deep-learning-based imaging and sensing techniques in complex environments. Also, I am interested in LLM-based multimodal computational sensing, and committed to applying multimodal LLM to computational sensing systems.

Selected Publications

[1] **Lintao Peng**, Chang Li, Wenhui Liu, Siyu Xie, Xue Chen, Fei Xiao*, Liheng Bian*, "Boosting bacterial detection with hyperspectral mining," ***Optica***, 2025, vol. 12, no. 3, pp. 315-324.

[2] **Lintao Peng**, Wenhui Liu, Siyu Xie, Lin Ye, Peng Ye, Fei Xiao, Liheng Bian*, "Uncertainty-driven parallel transformer-based segmentation for oral disease dataset," ***IEEE Transactions on Image Processing***, 2025, vol. 34, pp. 1632-1644.

[3] **Lintao Peng**, Chunli Zhu and Liheng Bian*, "U-shape transformer for underwater image enhancement", ***IEEE Transactions on Image Processing***, 32, 3066-3079 (2023). ESI highly cited paper, ESI hot paper

[4] **Lintao Peng**, Siyu Xie, Hui Lu, Liheng Bian*, "Large-scale single-pixel imaging and sensing," ***Advanced Photonics Nexus***, 2025, vol. 4, no. 2, pp. 026010.

[5] **Lintao Peng**, Liheng Bian*,"Adaptive dual-domain learning for underwater image enhancement," ***AAAI*** 39 (6), 6461-6469 (2025).

[6] **Lintao Peng**, Siyu Xie and Liheng Bian*, "Uncertainty-Driven Spectral Compressive Imaging with Spatial-Frequency Transformer", ***ECCV***, 54-70 (2024).

[7] **Lintao Peng,** Siyu Xie, Tong Qin, Lu Cao, and Liheng Bian*, "Image-free single-pixel object detection", ***Optics Letters***, 48, 2527-2530 (2023).

[8] **Lintao Peng**, Liheng Bian*, Tiexin Liu, and Jun Zhang, "Agile wide-field imaging with selective high resolution", ***Optics Express***, 29, 35602-35612 (2021).

[9] Liheng Bian*, Haozhe Song, **Lintao Peng**. *et al.* "High-resolution single-photon imaging with physics-informed deep learning", ***Nature Communications***, 14, 5902 (2023).

[10] Yongqi Cao, Hui Bao, **Lintao Peng**. *et al.* Single-Photon-Camera-Based Time and Spatially Resolved Electroluminescence Spectroscopy for Micro-LED Analysis. ***ACS Photonics***,2024, vol. 11, no. 2, pp. 731–736.

[11] Sitian Liu, **Lintao Peng** and Chunli Zhu*. "Adaptive dual-domain learning for hyperspectral anomaly detection with state space models." ***IEEE Transactions on Geoscience and Remote Sensing*** (2025).

[12] Sitian Liu, Chunli Zhu*, **Lintao Peng**. *et al.* "Wavelet-based diffusion with spatial-frequency attention for hyperspectral anomaly detection." ***International Journal of Applied Earth Observation and Geoinformation*** (2025),

142: 104662.

Academic Service

Journal Reviewer: IEEE TIP, IEEE TCSVT, IJCV, OE, OL, …

Conference Reviewer: CVPR, ECCV, NeurIPS, AAAI, ICCV, ACM MM, …

Awards & Honors

- Excellent Ph.D. Dissertation Seedling Fund of Beijing Institute of Technology (Top 0.5%) 2025
- National Scholarship for Ph.D. Students (Top 1%) 2024
- The Special Scholarship for Ph.D. Students of Beijing Institute of Technology (Top 5%) 2023&2025
- Excellent Graduation Thesis of Xidian University 2020
- First Prize in the China Undergraduate Physics Tournament (CUPT) 2018

Research Experiences

Physical Priors Enhanced Computational Imaging and Sensing

- End-to-end computational imaging networks typically learn a direct mapping from measurements to ground truth while neglecting physical priors of real scenes, leading to poor generalization. Non-end-to-end approaches, on the other hand, rely on iterative optimization, resulting in high computational cost and long reconstruction time. To address these issues, we propose a physics-prior-enhanced end-to-end computational imaging and sensing framework. Specifically, by analyzing physical imaging models, noise models, and statistical features of large-scale measurements, we extract effective physical priors to guide both reconstruction and sensing process. These priors are embedded into the network architecture and loss design, enabling efficient and high-quality end-to-end imaging and perception. We demonstrate the effectiveness of this approach in underwater imaging, spectral imaging, and scattering imaging, where extensive experiments show superior performance over existing methods with lower computation and memory cost.
- Five first-author papers have been accepted to Optica, TIP, ECCV, and AAAI, including an ESI highly cited paper and an ESI hot paper.

Megapixel Single-photon Imaging and Sensing under Extreme Low Light

- Conventional imaging devices fail in extreme low-light conditions, while SPAD cameras can detect weak light by amplifying single incident photons through the avalanche multiplication effect. However, they still suffer from low resolution and high noise. To address this, we first proposed a physics-based multiparametric noise model for SPAD cameras to generate large-scale training data. By integrating compressed sensing algorithms, we developed a megapixel single-photon imaging system that enhanced the resolution of a 32×32 SPAD camera to 1024×1024. Finally, leveraging the physical priors of the hardware system and SPAD measurements, I designed an end-to-end single-photon imaging network, pretrained on large-scale simulated data and fine-tuned with limited real data, achieving high-quality real-time megapixel single-photon imaging under extremely low light.
- One first-author papers have been accepted to Advanced Photonics Nexus, one first-author paper under review.

Computational Imaging and Sensing with Joint Encoder–Decoder Optimization

- In conventional computational imaging systems, hardware encoding and software decoding are designed independently, requiring manual parameter tuning for each use and adding significant computational overhead to reconstruction. To address this, we propose a encoding-decoding joint optimization strategy that embeds the physical imaging model and hardware parameters into a deep learning model, enabling joint optimization of hardware settings (e.g., modulation masks) and the reconstruction network during training. After training, the optimized parameters are used to build the hardware encoding system, which works seamlessly with the decoding network to achieve efficient and high-quality computational imaging and sensing. Experiments demonstrate that this approach not only achieves state-of-the-art imaging and sensing performance but also reduces hardware parameter complexity by two orders of magnitude.
- Two first-author paper have been accepted to Optics Express and Optics Letters, one first-author paper under review.